

GABRIELA AGUIRRE

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512-975-0045 | Austin, TX |

EDUCATION

The University of Texas at Austin, Austin, TX

Graduating August 2027

Bachelor of Science in Mechanical Engineering

Relevant Coursework: Chemical Process Safety, Heat Transfer, Mechatronics, Materials Engineering, Fluid Mechanics, Solids, Machine Elements, Corrosion, Powertrains, Dynamic Systems and Controls

EXPERIENCE

Texas Department of Transportation, *Engineering Intern*

May 2024-August 2024

- Shadowed a Civil Engineering Project Manager working on construction of the Walnut Creek Bridge expansion.
- Organized tickets/receipts from prior projects to make record-keeping easier.

Nuclear Engineering and Teaching Laboratory, *Desk Assistant*

January 2024-May 2024

- Budgeted for new office supplies and technology that were high quality and affordable
- Maintained front desk operations and appointments for employees
- Answered phone calls and prepared documents containing private government information

W3 Events, *Security Guard*

June 2024 - Present

- Managed crowd flow during 16-hour festival shifts with up to 50,000 attendees, establishing and holding clear access lanes so medical staff could reach patrons quickly.
- Directed emergency response in dense crowds, physically extracting unconscious attendees to safety and coordinating handoff to on-site EMS under time pressure.

PROJECTS

Kid Trax Dodge Charger Gear Box Restoration

June 2026 - July 2026

- Lead a team of two other people to repair a Kid Trax Dodge Charger. The battery cable was repaired, the clear plastic mold that covered the lights was 3D printed, and the gears were replaced from spur gears to helical gears to improve load bearing capacity.

AC/DC 3D-Printed Windmill

January 2026 - May 2026

- Designed and built a wind speed indicator with a partner. The generator's AC output was converted to DC with a bridge rectifier and smoothing capacitor, an LM339 quad comparator sorted the signal into low, medium, and high ranges using two tunable potentiometers, and the windmill housing and blades were 3D printed.

LEADERSHIP AND COMMUNITY INVOLVEMENT

Society of Hispanic Professional Engineers (SHPE), *SHPE Jr. Cochair*

August 2024 - May 2026

- Created and lead lesson plans to inspire high schoolers to pursue engineering and college
- Organized carpools for UT students mentoring high school students at Del Valle High School.
- Completes tasks such as acquiring drinks and food prior to general meetings to accommodate over 100 in attendance.

Sigma Lambda Alpha Sorority, Inc. (Señoritas Latinas en Acción),

January 2025 - Present

- Designed, manufactured and distributed over 200 stickers with visually engaging graphics to promote the sorority, increasing the recruitment rate by 20%

TECHNICAL SKILLS, CERTIFICATIONS

- Proficient in Solidworks, Matlab, AutoCAD, Adobe Apps; Familiar with Multisim and Revit
- OSHA 10-Hour Safety Training

HOBBIES AND INTERESTS

Outside of my role as a student, I enjoy volleyball, volunteering and listening to music. In my free time, I'm currently working on building a portable Raspberry Pi photobooth.